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Lab Report 2

Data Flow Diagram (DFD) Design

Project:

Nepali Lifestyle Based Diet and Fitness Advisory System

Course	Software Engineering (Practical)
Course Code	ENCT 352

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1. Introduction

A Data Flow Diagram (DFD) is a graphical representation that shows how data moves through a system, identifying the external entities that interact with the system, the processes that transform data, the data stores where information is retained, and the flows that connect them. Unlike the Software Requirements Specification, which describes *what* a system must do in textual form, a DFD models *how* data actually travels between the user and the internal processes, giving developers a clearer picture of system behaviour before design and coding begin.

DFDs are prepared after the Software Requirements Specification because the requirements first establish the functional scope of the system, and the DFD then translates that scope into a structured, visual data flow model. This model helps the development team identify the exact inputs, outputs, and internal processing steps required to satisfy the requirements, and it exposes any gaps or inconsistencies before implementation starts.

For the Nepali Lifestyle Based Diet and Fitness Advisory System, the DFDs prepared in this lab illustrate how user data, food images, and health goals flow between the user, the internal processes such as user management, food analysis, and recommendation generation, and the supporting data stores. These diagrams form the structural basis for the system design that follows in later phases of the project.

2. Data Flow Diagrams

This chapter presents the Context Diagram, the Level 1 DFD, and the Level 2 DFD developed for the Nepali Lifestyle Based Diet and Fitness Advisory System. Each diagram is described in terms of its external entities, processes, data stores, and data flows.

2.1 Context Diagram

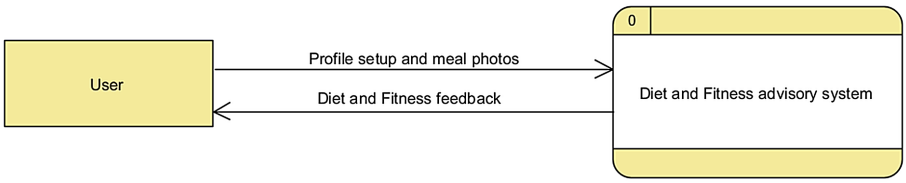


Figure 2.1: Context Diagram of the Nepali Lifestyle Based Diet and Fitness Advisory System

The Context Diagram shown in Figure 2.1 represents the highest level of the DFD, labelled as process 0. It provides an overview of the entire system as a single process and defines its boundary with the external environment.

The only external entity in this diagram is the **User**. Two data flows connect the User to the Diet and Fitness Advisory System: "Profile setup and meal photos", which carries the user's profile information and meal images into the system, and "Diet and Fitness feedback", which carries the generated recommendations back to the User. No data stores are shown at this level, since the Context Diagram treats the system as a single, undivided process.

As the first and highest-level diagram, the Context Diagram does not expand any previous level. It instead establishes the overall system boundary that the Level 1 DFD subsequently decomposes into individual processes.

2.2 Level 1 DFD

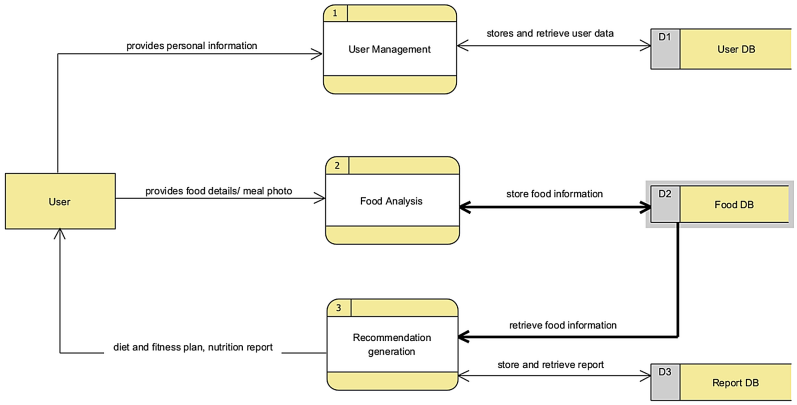


Figure 2.2: Level 1 DFD of the Nepali Lifestyle Based Diet and Fitness Advisory System

Figure 2.2 presents the Level 1 DFD, which decomposes the single process shown in the Context Diagram into three major processes of the system.

The external entity remains the **User**. The three processes at this level are **1 User Management**, **2 Food Analysis**, and **3 Recommendation Generation**. Three data stores support these processes: **D1 User DB**, **D2 Food DB**, and **D3 Report DB**. The User provides personal information to User Management, which stores and retrieves user data from the User DB. The User also provides food details or a meal photo to Food Analysis, which stores food information in the Food DB. Recommendation Generation retrieves food information from the Food DB, stores and retrieves reports from the Report DB, and finally sends the diet and fitness plan along with the nutrition report back to the User.

This diagram expands the single process of the Context Diagram into the three functional areas of the system: managing user profiles, analysing food, and generating recommendations. Each of these processes now has a defined data store and its own set of data flows, giving a clearer view of how the system separates its responsibilities internally.

2.3 Level 2 DFD

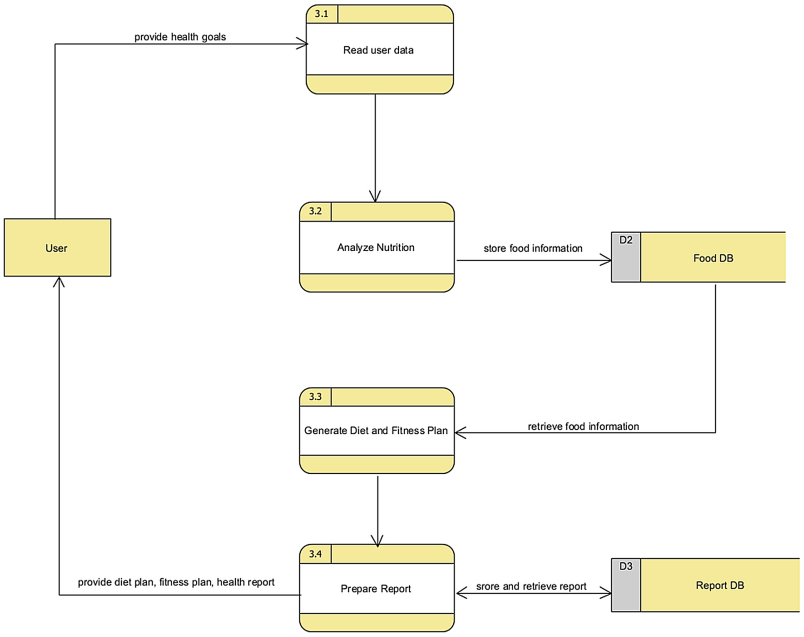


Figure 2.3: Level 2 DFD - Decomposition of Process 3 (Recommendation Generation)

Figure 2.3 shows the Level 2 DFD, which further decomposes process 3, Recommendation Generation, from the Level 1 DFD into four sub-processes.

The external entity is again the **User**. The four sub-processes are **3.1 Read User Data**, **3.2 Analyze Nutrition**, **3.3 Generate Diet and Fitness Plan**, and **3.4 Prepare Report**. Two data stores appear at this level, the **Food DB** and the **Report DB**. The User provides health goals to Read User Data, which passes this information to Analyze Nutrition. Analyze Nutrition stores food information in the Food DB, which is later retrieved by Generate Diet and Fitness Plan. The output of this sub-process flows into Prepare Report, which stores and retrieves report data from the Report DB and finally provides the diet plan, fitness plan, and health report back to the User.

This diagram refines process 3 of the Level 1 DFD by breaking down Recommendation Generation into a sequential set of sub-processes that read user health goals, analyse nutrition against stored food information, generate the actual plan, and prepare the final report. It shows the internal working of the recommendation process without altering the data flows already defined at the higher level.

3. Overall Discussion

The three diagrams developed in this lab show a clear progression from a general overview to a detailed internal structure. The Context Diagram treats the Nepali Lifestyle Based Diet and Fitness Advisory System as a single process interacting with the User, establishing the system boundary. The Level 1 DFD then breaks this single process into three functional areas, User Management, Food Analysis, and Recommendation Generation, each supported by its own data store. The Level 2 DFD goes further by decomposing Recommendation Generation into four sequential sub-processes, revealing how user health goals are read, analysed against stored food data, and converted into a diet and fitness plan.

Across these levels, information consistently flows from the User into the system, through the relevant process, into or out of a data store, and back to the User as feedback. This layered structure mirrors the system architecture described in the SRS, where profile management, food analysis, and recommendation generation form distinct but connected modules built on a relational data layer.

These diagrams directly support the functional requirements defined in Lab 1, such as profile management, meal logging, and personalized advisory generation. By visualizing the exact data flows before implementation, the development team can verify that every requirement has a corresponding process and data path, making the DFDs a valuable reference during the design and coding phases that follow.

4. Conclusion

In this lab, the Context Diagram, Level 1 DFD, and Level 2 DFD were developed for the Nepali Lifestyle Based Diet and Fitness Advisory System, based directly on the requirements identified in Lab 1. These diagrams illustrate how user information, meal data, and health goals move through the system's processes and data stores at increasing levels of detail.

Data Flow Diagrams are important because they translate textual requirements into a visual model of system behaviour, making it easier to identify processes, data stores, and flows that must be implemented. Working across multiple levels, from the Context Diagram down to the Level 2 decomposition, allows the team to understand both the overall system boundary and the detailed internal workings of specific processes such as recommendation generation.

Completing this lab has clarified how the major modules of the system interact and exchange data, providing a solid foundation for the system design phase. Lab 2 therefore prepares the project for the next phase of development, in which these data flows will be translated into concrete database schemas and module-level designs.